

KOMAL DISANAYAKA

AI FULL STACK DEVELOPER

✉ komaldissanayaka.work@gmail.com ☎ +94 71 341 9970  [Komal Disanayaka](#)  [Komal Disanayaka](#)

SUMMARY

I'm a passionate learner and builder who loves exploring how Artificial Intelligence and modern web technologies can be used to solve real-world problems. I enjoy designing clean, scalable, and user-focused digital solutions—from intelligent AI systems to full-stack web applications.

I work with Machine Learning and Deep Learning models to build smart, data-driven systems. I'm especially interested in applying AI to automation, prediction, and decision-making to create meaningful impact.

I develop end-to-end applications using modern technologies, focusing on performance, usability, and maintainable code. I enjoy turning ideas into fully functional products. Continuously learning and challenging myself to grow every day.

EDUCATION

Sri Lanaka Institute of Information Technology | Malabe, Sri Lanka

BSc (Hons) in Data Science (2024 july - Present) | Graduation Year - 2028
Cumulative GPA : 3.73

G.C.E. Advanced Level - Science Stream (Mathematics) | 2023(2024)

Successfully completed with 2C and 1S.

G.C.E. Ordinary Level | 2020(2021)

Successfully completed with 8A and 1C.

DEVELOPMENT EXPERIENCE

- Worked in Agile (Scrum) development environments, participating in sprint planning, daily stand-ups, sprint reviews, and retrospective meetings.
- Collaborated with cross-functional teams to design, develop, test, and deploy full-stack web, mobile, AI, and IoT applications.
- Used Git and GitHub for version control, feature branching, pull requests, code reviews, and team collaboration.
- Followed RESTful API development principles and integrated frontend, backend, databases, and AI services.
- Applied Object-Oriented Programming (OOP) principles and clean coding practices to build scalable and maintainable software.
- Participated in software testing, debugging, issue tracking, and performance optimization throughout the development lifecycle.
- Experienced in deploying applications using modern cloud platforms such as Vercel, Render, and Hugging Face Spaces.

CERTIFICATE

1. Dean's List – SLIIT

- Issued by: Sri Lanka institute of information technology
- Date: June 2026

Description: Certificate of Excellence in reorganization of academic excellence in the 2nd year 1st Semester.

PROJECTS

01 . Smart Water Leak & Quality Detection System – IoT Embedded System

- Objective: Developed an IoT-based smart water management system for real-time water level monitoring, leak detection, automatic valve control, and water quality analysis using embedded sensors.
- Technologies Used: Arduino, Embedded C/C++, Arduino IDE, HC-SR04 Ultrasonic Sensor, YF-S201 Water Flow Sensors, pH Sensor, HC-05 Bluetooth Module, Solenoid Valve, Serial Communication, Interrupts, IoT.
- Result: Achieved near real-time leak detection, reduced water wastage by 30–40%, decreased manual monitoring by 80%+, and delivered a low-cost IoT solution (~LKR 15,000) with automated water flow control.

02 . Face Recognition AI Web Application – AI/ML Web Application

- Objective: Developed a full-stack AI web application for real-time face detection and recognition with an intuitive image upload interface.
- Technologies Used: Python, Flask, OpenCV, OpenFace, Caffe SSD (ResNet-10), Scikit-learn, SVM, NumPy, HTML, CSS, JavaScript.
- Result: Achieved fast and accurate multi-face recognition using 128-dimensional facial embeddings, confidence-based SVM classification, and automatic face annotation with an extensible AI pipeline.

03 . PetBuddy – Smart Pet Care Management Mobile Application

- Objective: Developed a full-stack mobile application for comprehensive pet care management, enabling pet profile management, health tracking, service booking, social interaction, and AI-powered pet care assistance.
- Technologies Used: React Native, Expo, Node.js, Express.js, MongoDB Atlas, JWT Authentication, Cloudinary, OpenRouter API, JavaScript.
- Result: Built a scalable cross-platform mobile application featuring secure authentication, digital health records, online service booking, community engagement, and an AI-powered virtual pet care assistant for an enhanced user experience.

04 . Real-Time Character Recognition System – Deep Learning Web Application

- Objective: Developed an end-to-end AI web application for real-time handwritten character recognition using a fine-tuned ResNet-18 deep learning model.
- Technologies Used: Python, PyTorch, Torchvision (ResNet-18), Flask, OpenCV, NumPy, HTML5 Canvas, CSS, JavaScript, CUDA, cuDNN.
- Result: Achieved high recognition accuracy through transfer learning, real-time image preprocessing, and GPU-accelerated inference, enabling fast and interactive handwritten character prediction via a web interface.

05 . Warehouse Safety Monitoring System – AI-Powered PPE Detection Web Application

- Objective: Developed a full-stack AI-powered web application for real-time warehouse safety monitoring by detecting Personal Protective Equipment (PPE) compliance from CCTV, webcam, image, and video streams.
- Technologies Used: React.js, Vite, Material UI, Recharts, Node.js, Express.js, MongoDB Atlas, Python, Flask, YOLOv8, OpenCV, Docker, JWT Authentication, Vercel, Render, Hugging Face Spaces.
- Result: Built a production-ready AI system capable of real-time detection of Helmet, No-Helmet, Vest, No-Vest, and Person with automated violation alerts, live compliance analytics, and role-based management, significantly improving workplace safety monitoring.

SKILLS

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|-------------------------|----------------|--------------------|
| • Deep Learning | • React | • Problem solving, |
| • Machine Learning | • Node JS | • Team working |
| • Java (OOP) | • MongoDB | • Yolo V8 |
| • Python Advance | • Git & GitHub | • Postman |
| • JavaScript/TypeScript | • Express JS | • Render |
| • React Native | • PostgreSQL | • Versal |
| • Rest API/Flask API | • Next JS | • Hugging Face |